Exercises

Exercise 7.1 · Euclid's algorithm. Find $\gcd(1{,}071, 462)$ with Euclid's algorithm, then simplify the fraction $\dfrac{462}{1{,}071}$.

Exercise 7.2 · Bézout pair. Find a pair of integers (u, v) such that $17u + 12v = 1$.

Exercise 7.3 · Coprime integers. Let n be a natural number. Prove that n and $n + 1$ are coprime, then that $2n + 1$ and $3n + 1$ are coprime too.

Exercise 7.4 · A homogeneous equation. Find all pairs of integers (x, y) such that $15x = 21y$.

Exercise 7.5 · Exercise books and pens. A bookshop sells exercise books at 1,100 ariary and pens at 700 ariary. A customer spends exactly 10,000 ariary. How many exercise books and pens did they buy?

Exercise 7.6 · Inverse modulo 26.

1. Explain why 7 has an inverse modulo 26 and check that 15 is one.
2. Solve $7x + 3 \equiv 10 \pmod{26}$.

Exercise 7.7 · A gcd that depends on n. Let n be a natural number and $d = \gcd(n + 5, 2n + 3)$.

1. Prove that d divides 7.
2. Find the integers n for which $d = 7$.